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Electromigration reliability of Cu₃Sn microbumps for 3D heterogeneous integration

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Abstract—3D heterogeneous integration (HI) and advanced packaging (AP) technologies require small volume, high-density interconnects for stacking discrete chips for which the reliability of interconnects becomes crucial. Intermetallic compounds (IMCs) based μ bumps have been shown to outperform solder-based μ bumps concerning their resistance to electromigration (EM) related failures, which is a key index to assess the interconnect reliability. Cu-Sn solid-liquid interdiffusion (SLID) bonding is an attractive low-cost wafer-level bonding technology for rapid manufacturing of full Cu₃Sn IMC μ bumps. However, SLID requires melting of Sn during the bonding process which poses risks and design challenges in manufacturing. Due to Sn squeeze-out during the bonding process, Sn melt could react with redistribution layers (RDLs) or metallization layers and form IMCs at undesired locations resulting in early failures thereby compromising the reliability. The Sn-squeeze out issue during bonding is addressed in this work by designing test structures with equal and unequal lateral dimensions of μ bumps in the top and bottom wafers. The effects of Sn-squeeze out on the EM resistance and reliability are compared in both designs. Significant improvement in the Sn-squeeze out and corresponding EM resistance was observed in the test structures manufactured with unequal lateral dimensions of μ bumps in the top and bottom wafers. FE element simulations were carried out to gain insights and assess the impact of Sn squeeze-out on the reliability and functionality of the Cu₃Sn μ bumps.

Keywords—Cu-Sn SLID bonding, electromigration, failure analysis, FE simulations, heterogeneous integration

I. INTRODUCTION

The slowdown of Moore's law has sparked abundant interest recently in "Advanced Packaging" technologies to realize multichip packaging for enhanced system performance, improved form factor, lower cost, and possibility of integrating chips (such as logic, memory, and ASIC) manufactured in different technology nodes in a single package [1]. Moreover, the ever-increasing requirements for enhanced functionalities on a single chip apart from CMOS, such as RF, MEMS, and optical, have brought forward the importance of 3D heterogeneous integration (3D HI) [2]. One of the key requirements in realizing the above goals is on fine-pitch high-density interconnects (HDI) with high bandwidth, low latency, enhanced reliability, and low cost to integrate or stack different chips in a package. The interconnects therefore

have a major role in the overall power, performance, area, and cost (PPAC) considerations of electronic chips.

Electromigration (EM) is a major well-known reliability concern in the interconnects where the atoms migrate due to momentum transfer from electrons under the influence of an electric field. The migration, in turn, could create voids (at the cathode) and hillocks (at the anode) in the interconnects resulting in open or short failures respectively. Combined with joule heating, the situation could get worse with failures related to thermomigration, and even thermal runaway [3]. Therefore, EM assessment of interconnects in 2.5D/3D package architectures becomes significant with the ongoing efforts of scaling them down to achieve HDI, which in turn could also result in current densities exceeding 1×10^5 A/cm².

Today hybrid bonds with dimensions beyond 1 μ m are possible for HDI [4-5]. However, manufacturing those is challenging because of stringent processes involved and is therefore mostly limited to chip foundries capable of operating in the front-end technology nodes [5]. On the other hand, flip chip (FC) solder μ bumps are limited to 20 – 30 μ m in diameter [6-7]. The FC technology is low-cost compared to hybrid bonds and is therefore utilized by both front-end foundries and back-end outsourced semiconductor assembly and testing (OSAT) companies. As the dimensions of FC solder bumps reduce, a major fraction of the solder volume is composed of intermetallic compounds (IMCs) [8-9]. The formation of IMCs in a solder volume presents reliability challenges concerning EM because of flux divergence at the IMC/solder interface [10]. In contrast, the EM reliability of full IMC μ bumps outperforms those of FC solder μ bumps [8,11].

Cu-Sn solid liquid interdiffusion (SLID) bonding is an attractive low-cost wafer-level bonding technology in which the μ bumps are fully composed of IMCs. Depending on the thickness of Sn (≈ 2 μ m) and Cu (> 4 μ m), full formation of Cu₃Sn IMC (M.P. – 676 °C) which is a thermodynamically stable phase could be obtained for a hold time duration of 1 hour during the bonding process at 320 °C [9]. Cu₃Sn μ bumps with dimensions as low as 5 μ m have been demonstrated recently [12]. The capability of manufacturing the Cu₃Sn μ bumps at wafer-level makes it an appealing low-cost alternative with dimensions below what FC could offer and with high EM resistance. The higher melting point of Cu₃Sn also makes possible stacking multiple chips desirable in 3D

HI without affecting the individual chips at the bonding process temperature ($\approx 300^\circ\text{C}$).

The EM reliability of Cu_3Sn IMC μ bumps therefore demands in-depth investigation. Limited studies are present in the literature concerning specifically the EM reliability of Cu_3Sn μ bumps in the past. In [8], no failures were observed in Cu_3Sn layer after stressing it for 5000 hours at a current density of $2.1 \times 10^5 \text{ A/cm}^2$ and temperature of 170°C . However, a 4% resistance increase in one of the samples stressed at a current density of $1.4 \times 10^5 \text{ A/cm}^2$ at 170°C was attributed to the damage in the Al metallization layer. In [11], similar observations of damage in Al-Cu metallization layer were made by stressing Cu_3Sn IMC μ bumps at a current density of $4 \times 10^5 \text{ A/cm}^2$ at a temperature of 200°C for 1000 hours. The Cu_3Sn IMC μ bumps were manufactured in the above studies by thermal annealing of FC μ bumps. In our earlier work [9], test structures were fabricated with Cu_3Sn μ bumps demonstrating wafer-level Sn-Sn SLID bonding. Since, SLID requires melting of Sn during the bonding process, Cu_3Sn was formed at undesired locations in the Cu metallization layer due to Sn-squeeze out. Because of that, failures such as microstructural changes due to thermal runaway and burnout of Cu_3Sn μ bumps were observed. That highlighted the inherent design limitations in realizing Cu_3Sn μ bumps carried out *via* wafer-level bonding.

In this work, to overcome the issue of Sn-squeeze out, the design of the test structures is revisited. With the new design, significant improvement in the test structures with smaller μ bumps were observed. The samples were current stressed at accelerated conditions to analyze the failures. Non-destructive techniques, such as lock-in thermography (LIT) and time domain reflectometry (TDR) were employed to locate the failure locations followed by destructive analysis for microstructural investigation utilizing scanning electron microscopy (SEM) and energy-dispersive spectroscopy (EDS).

II. EXPERIMENTAL DETAILS

A. Design and fabrication of test structures

Different types of test structures (TS), such as 2-bump, daisy chain and kelvin were designed and incorporated in a 4-inch mask process. In this work, only 2-bump TS were investigated. Fig. 1(a) shows the schematic image of a 2-bump test structure. It was designed and incorporated into the mask with different pitch sizes. The μ bumps in the TS were square in shape with length (L) and width (W) equal. In this work, the Sn squeeze-out issue was addressed by deliberating design considerations with unequal lateral dimensions of top and bottom μ bumps. The TS are classified in two ways: (i) with equal lateral dimension of top and bottom μ bump ($L_{\text{TOP}} = L_{\text{BOT}}$), batch-1, Fig. 1(b), and (ii) reduced dimension of top μ bump ($L_{\text{TOP}} < L_{\text{BOT}}$), batch-2, Fig. 1(c). The batch-1 TS were from the earlier work and is utilized in this work as a reference [9]. In batch-2, the lateral dimensions of the μ bumps on the top wafer was reduced over a range of $5 - 20 \mu\text{m}$ compared to the μ bump dimensions on the bottom wafer as shown in Table 1. Wafer level bonding was carried out on a pair of 4-inch silicon wafers after patterning the test structures on both top and bottom silicon wafers with subsequent electroplating of Cu and Sn. The bonding was carried out at 320°C with a hold time of 1 hour which results in a complete formation of Cu_3Sn IMCs in the μ bumps of the TS. The detailed fabrication process flow of the TS and process

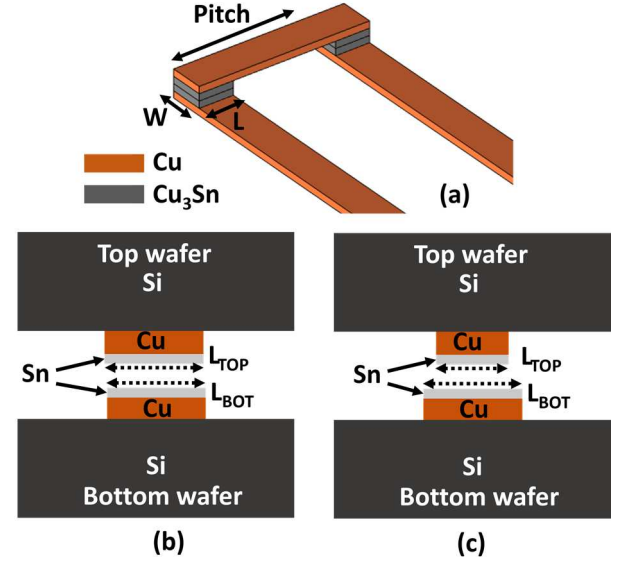


Fig. 1. (a) Schematic image of a 2-bump test structure, and schematic image of μ bumps from (b) batch-1 ($L_{\text{TOP}} = L_{\text{BOT}}$), and (c) batch-2 ($L_{\text{TOP}} < L_{\text{BOT}}$).

parameters could be referred from the earlier work [9]. With the same bonding force (20 MPa), bonding temperature (320°C) and thickness of electroplated Cu ($\approx 5 - 5.5 \mu\text{m}$) and Sn ($\approx 2 - 2.5 \mu\text{m}$), there was a significant improvement in the TS from batch-2 for smaller sized μ bumps with regards to the Sn-squeeze-out as will be seen later in Sec IIIA.

TABLE 1. DIMENSIONS OF THE BOTTOM AND TOP μ BUMPS IN BATCH-2

Bottom bump, $L_{\text{BOT}} = W_{\text{BOT}} (\mu\text{m})$	Top bump, $L_{\text{TOP}} = W_{\text{TOP}} (\mu\text{m})$
25	20
35	25
50	40
50	30

B. EM Tests

The EM reliability tests of TS from batch-1 were carried out with Keithley 2231A-30-3 three-channel dc power supply with chips placed on a thermal chuck. The EM reliability tests of TS from batch-2 were carried out on a Micromanipulator manual probe station employing Nextron micro probe system (MPS) as a thermal chuck. The TS from batch-2 were stressed using B1500 semiconductor parameter analyzer from Keysight Technologies. All the EM reliability tests were carried out at an elevated temperature of 150°C with nominal current densities ranging from $5 \times 10^4 \text{ A/cm}^2$ to $1.28 \times 10^5 \text{ A/cm}^2$. The current densities were calculated based on the ideal cross-sectional dimensions of the μ bumps in the bottom wafer (for e.g., in batch-2: $25 \mu\text{m}^2$, $35 \mu\text{m}^2$ and $50 \mu\text{m}^2$).

C. Non-destructive evaluation

To monitor the formation and evolution of EM related failures in the TS, a combination of non-destructive techniques, i.e., time domain reflectometry (TDR) and lock-in thermography (LIT) was employed. Monitoring the impedance spectrum of the TS before and after EM stress tests with TDR yields information on the origin and location of failures in the TS. The impedance spectrum of the TS was recorded before and after the EM stress tests for comparison.

For continuation of the EM stress test conditions and to induce forthcoming of the migration process, the sample was placed on a micro heater set to 150 °C with an integrated PT100. The heater was controlled by a power supply (Gossen Metrawatt LSP) and the temperature was measured employing a Fluke 8846A. The TS were electrically contacted using a RF probe (Picoprobe 40A-GS) which was fixed onto a micromanipulator (Formfactor DPP 210) for precise xyz-positioning of the tip. For inducing electrical (DC) stress to the TS and to simultaneously measure the electrical impedance spectrum of the TS, the prober was connected to a three port L-C-network (bias-tee, mini circuits). The DC-bias current was applied using a (Keithley Instruments 2601A) and the acquisition of the impedance spectrum was performed using a network analyzer (Copper Mountain Planar 304/1). Upon the recognition of progressive characteristic alteration in the impedance spectra, non-destructive LIT was conducted. LIT is capable of visualizing thermal emissions in response to a characteristic electrical excitation of the sample and were conducted using an IR-sensitive camera (IRCAM Velox 327k camera) equipped with either a 2.5x or 5x magnification lens.

D. Destructive failure analysis

Destructive analysis was carried out to prepare the cross-section for evaluating the failure modes and microstructural changes in the TS after EM tests. The cross-sections of the TS were prepared by cutting the chips in two halves at the vicinity ($\approx 300 \mu\text{m}$ away) of the μbumps by employing a femtosecond laser micromachining tool. Subsequently, the chips were molded in epoxy and cured followed by grinding and polishing steps employing standard metallographic methods to prepare the cross-section for subsequent SEM and EDS analysis.

E. SEM and EDS Characterization

SEM characterization and elemental mapping of the cross-sections were carried out with a dual beam focused ion beam (FIB)-SEM JEOL JIB-4700F equipped with the Oxford instruments Ultim Max 100 EDS detector.

III. RESULTS AND DISCUSSION

A. SEM and cross-sectional analysis of virgin samples

The improvement in the Sn-squeeze out because of the new design of the test structure was evaluated. Virgin samples which were as fabricated (unstressed) were prepared for cross-sectional analysis and examination. Fig. 2(a) shows the cross-section image of the TS from batch-1, where Cu_3Sn IMC formations at unwanted locations were observed due to Sn-squeeze out during the bonding process [9]. Those locations are demonstrated with yellow circles. The Cu track connecting the two μbumps should be uninterrupted as shown in the schematic of Fig. 1(a). In contrast, Sn-squeeze out is significantly reduced in the TS with bottom and top μbump dimensions of 25 μm and 20 μm respectively from batch-2. Fig. 2(b) shows the SEM micrograph and EDS elemental mapping of Cu, Sn, and Si. Fig. 2(c) shows their respective atomic percentage plots across the cut line location shown in the inset, and Cu_3Sn IMC was confirmed with atomic weight percentage of Cu – $75.4\% \pm 1.6\%$ and Sn – $24.0\% \pm 1.5\%$. However, in batch-2 for larger sized μbumps (bottom μbump), i.e., 35 μm and 50 μm , formation of Cu_3Sn in the Cu track due to the squeeze out was observed which is due to the increase in the Sn volume. Fig. 2(d) shows the SEM cross-section of a 35 μm device where Cu_3Sn formation in the upper Cu trace

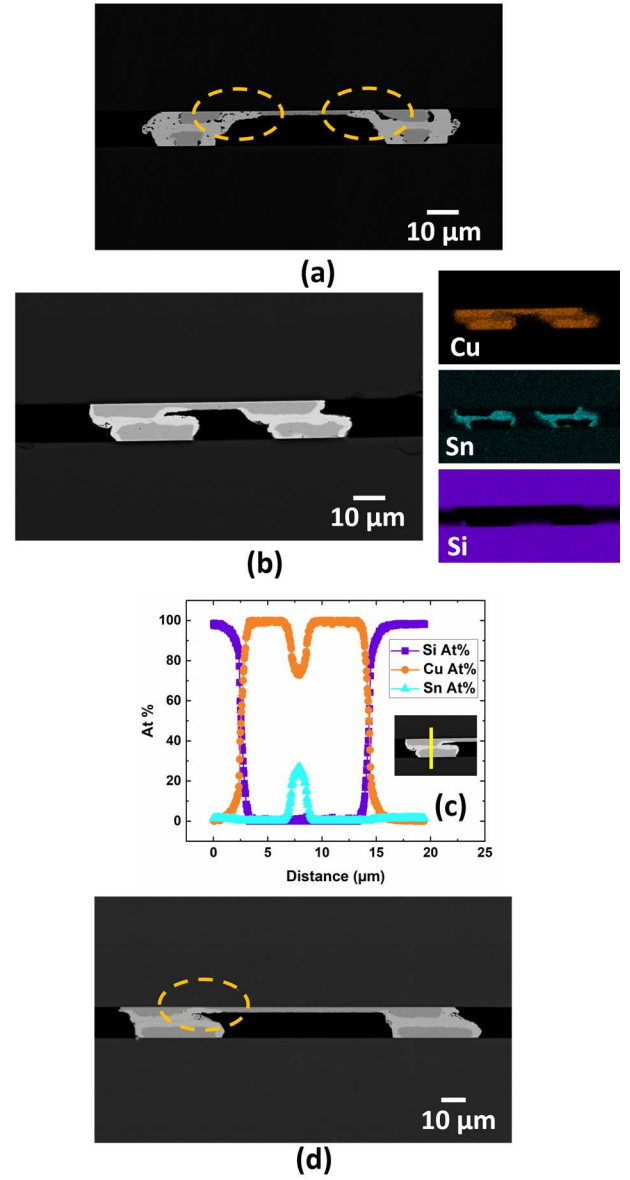


Fig. 2. (a) SEM image of 25 μm batch-1 test structure [9], (b) SEM image and EDS elemental maps of 25 μm batch-2 test structure, (c) atomic percentage versus distance plot across the cutline shown in the inset, and (d) SEM image of 35 μm test structure from batch-2 where Cu_3Sn was observed near the left μbump at the upper Cu trace.

layer near to the left μbump is observed which is marked in yellow circle. Because of this reason, only 25 μm (with top μbump dimension 20 μm) test structures from batch-2 were stressed to evaluate the EM reliability of the improved design.

B. EM test results

The EM reliability of the Cu_3Sn μbumps and the corresponding failure time were monitored for samples from both batches. The failure criteria in the test bench were set to 100% of initial resistance increase. Fig. 3(a) shows the EM test result of 25 μm TS from batch-1 (25 μm _TS1_B1) which failed after 142 hours of current stressing at a current density of $\approx 1 \times 10^5 \text{ A/cm}^2$. Fig. 3(b) shows the EM test result of 50 μm TS from batch-1 (50 μm _TS2_B1). The device failed after 37 hours of stressing at a current density of $\approx 5 \times 10^4 \text{ A/cm}^2$. Fig. 3 (c-d) shows the EM test result of 25 μm TS from batch-2 named 25 μm _TS1_B2 and 25 μm _TS2_B2 respectively. Significant improvement in the EM reliability

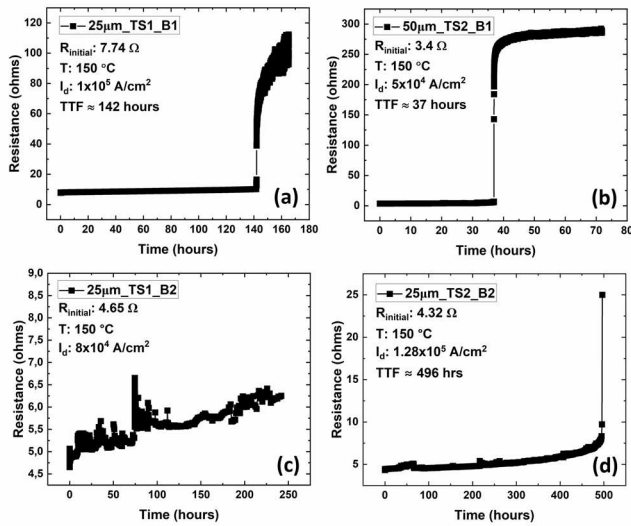


Fig. 3. EM test results of samples from batch-1, (a) 25μm_TS1_B1, (b) 50μm_TS2_B1, and batch-2, (c) 25μm_TS1_B2, (d) 25μm_TS2_B2.

was observed based on the EM stress tests. The 25μm_TS1_B2 was stressed at a current density of $\approx 8 \times 10^4$ A/cm². After stressing the sample for 242 hours, the failure criteria of 100 % resistance increase were not met and the test was stopped, Fig. 3(c). Fig. 3(d) shows the EM test result of 25μm_TS2_B2 which was stressed at a current density of $\approx 1.28 \times 10^5$ A/cm². The tests ran for ≈ 496 hours until the failure criteria of 100% resistance increase was met. However, significant oxidation and degradation of the Cu contact pads were observed in this sample and will be discussed in detail in subsequent Sec IIIC.

C. Non-destructive failure analysis

The non-destructive failure analysis of the batch-2 TS indicated degradation in the Cu traces and contact pads, but not in Cu₃Sn IMC μbumps. The EM stress test was carried out in ambient at 150 °C which also led to the oxidation of the Cu contact pads resulting in an increased contact resistance. Therefore, the resistance increase observed in Fig. 3(d) for 25μm_TS2_B2 after EM stress test is likely dominated by the degradation in the Cu contact pads and traces. For investigating this hypothesis, both LIT inspection and TDR measurements were carried out on 25μm_TS2_B2 before and after EM stress test. Since Si is partially transparent in the infrared band, non-destructive inspection of the interface between the two Si-dies is possible. Fig. 4 shows the LIT amplitude image of 25μm_TS2_B2 before EM stress test superimposed on the IR-emissivity image of the bonding interface. The red-framed inlet shows the pure emission contrast and allows inspection of the integrity of the Cu traces in virgin (unstressed) state. On the other hand, Fig. 5 shows the results of the analysis of 25μm_TS2_B2 sample after EM stress tests. Fig. 5(a) contains the LIT amplitude image superimposed on to the emissivity image of the sample. The bright spot in the red-framed area shows thermal emissions following the electrical excitation indicating high ohmic resistivity. Fig. 5(b) shows LIT-amplitude image of the red-framed area in Fig. 5(a), however, recorded with a higher magnifying objective. Fig. 5(c) shows the results of an IR-microscopy inspection only containing the emissivity contrast. The Cu trace shows degradation in the area where the LIT amplitudes are high. Since LIT visualizes thermal emissions in response to an electrical excitation, the bright

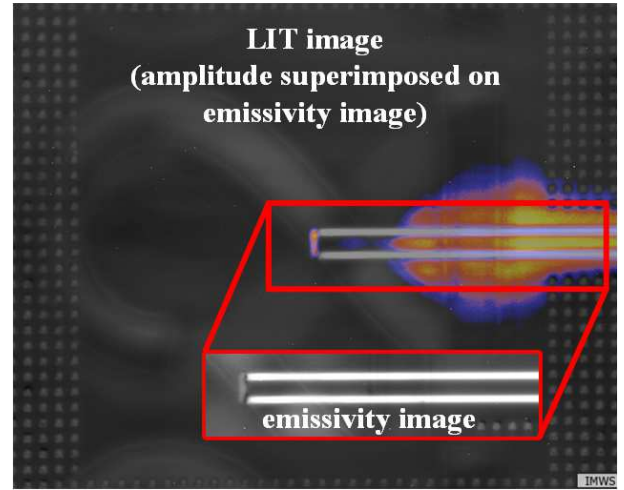


Fig. 4. LIT- and IR-image of 25μm_TS2_B2 sample before EM stress test. The outer image shows the LIT amplitudes superimposed onto the emissivity image of the samples interface. The red-framed inlet entitled “emissivity image” contains a higher resolution IR-image of the metal lines. Prior to stressing the metal lines are intact and show no indication of degradation.

spot is expected to indicate a degradation related to increase in the electrical resistivity as was observed in Fig. 3(d). The optical microscopy image of the contact pads is shown in Fig. 5(d) representing the orange-framed area in Fig. 5(a). Profound degradation of the pads as indicated by the green arrows is evident. Furthermore, an electrically conductive path bridging the two pads was formed under the stress condition and is indicated by the red arrow. Fig. 6 shows the impedance spectrum of 25μm_TS2_B2 sample measured from the TDR analysis before and after EM stress test. The graph contains the electrical response to a broad band step function. While the unstressed state shows a decreasing impedance, the curve recorded after EM stress test indicates a high impedance, supporting the above-described hypothesis. Since the TDR equipment has been calibrated such that 0 ns coincides with the end of the probe tip, the high reflectivity already occurs at the interface between the probe tip and the Cu contact pad.

D. Destructive failure analysis

Fig. 7(a) shows the cross-section of the sample 25μm_TS1_B1 after the EM stress test. As can be seen there was a burnout with complete destruction of the TS. Fig. 7(b) shows the cross-section of 50μm_TS2_B1. Fig. 7(c) shows the enlarged image of the left μbump with the corresponding EDS elemental maps of Cu, Sn, and Si. Fig. 7(d-e) shows the atomic percentage of Cu and Sn across two different cut line locations shown in the inset. Clearly, there was microstructural changes in the Cu₃Sn IMC in the squeeze-out region because of thermal runaway. These kinds of failures were observed and reported before [9].

Fig. 8(a) shows the cross-section of the sample 25μm_TS1_B2 from batch-2 with the corresponding EDS elemental maps of Cu, Sn and Si. No microstructural changes were observed and the μbumps were intact with Cu₃Sn IMC. This was confirmed by the atomic percentage plots of Cu – $72.7\% \pm 1.6\%$ and Sn – $25.9\% \pm 1.6\%$ across the cut line shown in the inset of Fig. 8(b). Similar observations were also made from the other sample, 25μm_TS2_B2, Fig. 8(c-d) with atomic percentage of Cu and Sn as $74.0\% \pm 1.2\%$ and $25.4\% \pm 1.2\%$ respectively. Therefore, the resistance increase in the

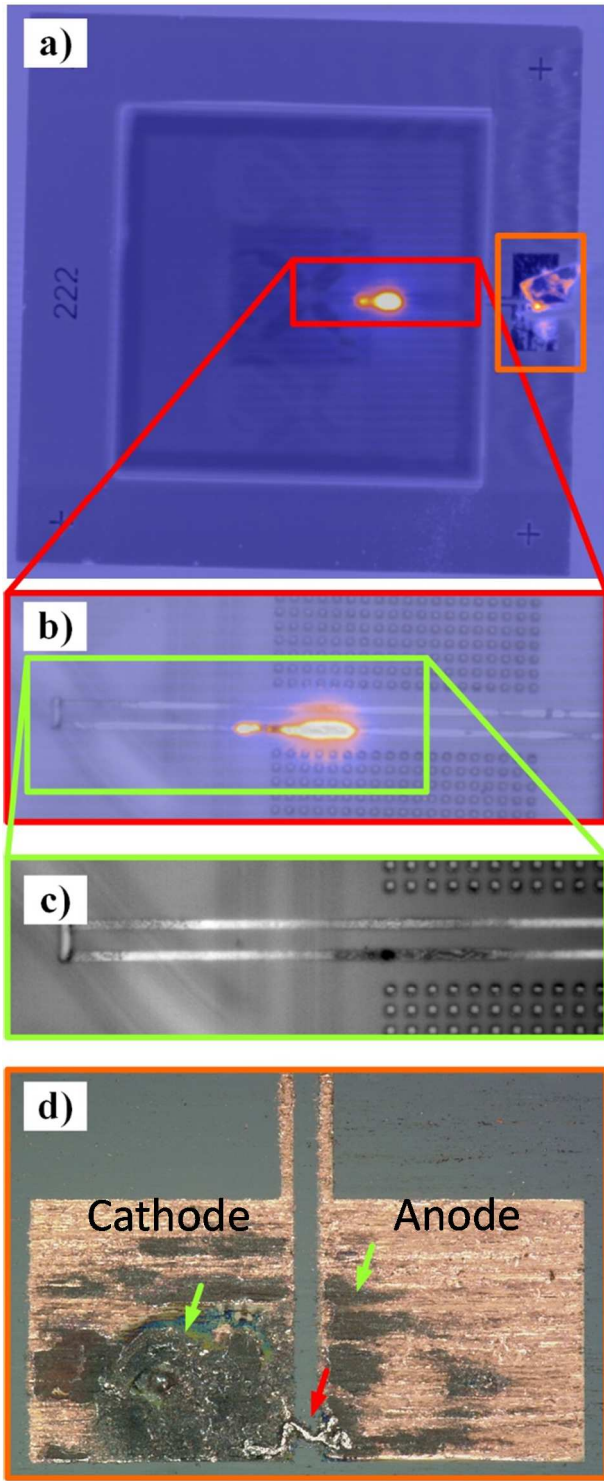


Fig. 5. LIT-, IR- and optical images of sample 25 μ m_TS2_B2 after EM stress test. (a) LIT-amplitude image superimposed on emissivity image of the samples interface between the two Si-dies. The red-framed area in (a) contains the Cu tracks. The orange-framed region contains the Cu contact-pads. High intensity in the false colors indicate thermal emission in response to the electrical excitation. (b) Higher magnification LIT-amplitude image of the red-framed inlet in (a). (c) IR-image of the green-framed inlet in (b). The dark regions in the Cu track indicate degradation explaining the thermal emission in the images in (a) and (b). (d) Optical image of the contact pads (orange framed area in (a)) exhibiting profound degradation (green arrows) and a bridge consisting of potentially migrated Cu shorting the two pads (red arrow).

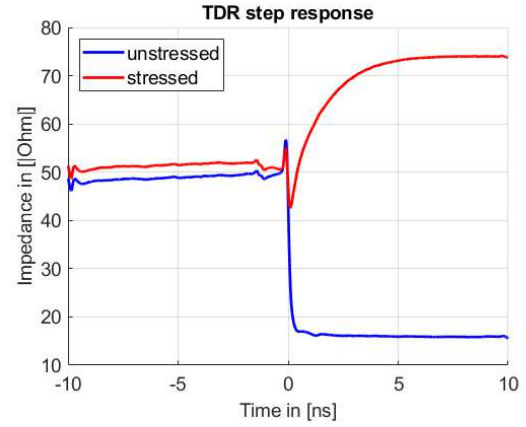


Fig. 6. TDR trace recorded in response to a step-function from the unstressed (blue) and the stressed (red) sample 25 μ m_TS2_B2. While the blue trace indicates a low impedance the red curve represents an increased line impedance likely dominated by the oxidation-affected contact resistance at the interface between tip and Cu contact pad.

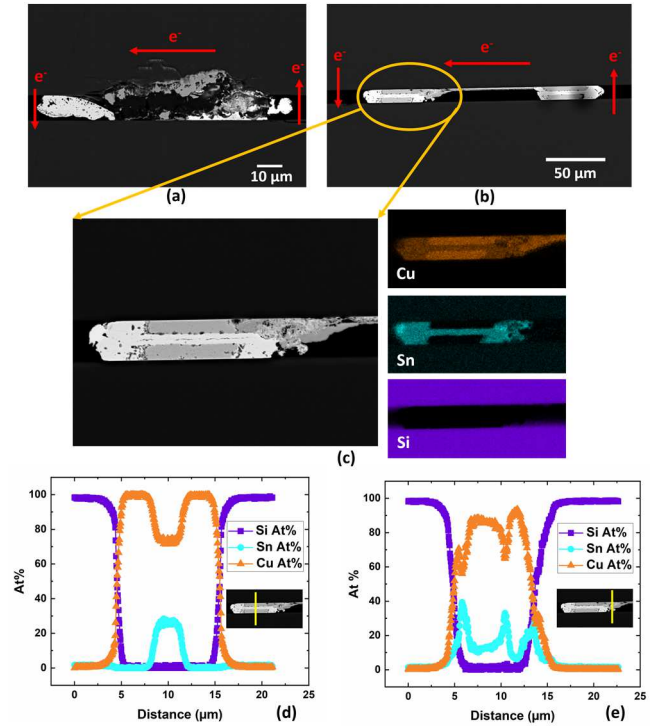


Fig. 7. (a) SEM image of 25 μ m_TS1_B1 after the EM test. (b) SEM image and elemental maps of 50 μ m_TS2_B1 after the EM test, (c) zoomed-in image with corresponding EDS elemental maps and (d-e) atomic percentage versus distance plots of 50 μ m_TS2_B1 after the EM test across different locations shown in the inset.

EM test result of 25 μ m_TS2_B2 shown in Fig. 3(d) was due to the oxidation and degradation of the Cu contact pads during the EM tests. The samples from batch-2 were stressed at higher current densities and for longer times than the samples from batch-1, but no failures and microstructural changes in the Cu₃Sn μ bumps were observed.

This demonstrates that the design of Cu₃Sn IMC μ bumps is crucial in overcoming Sn-squeeze out issue and therefore avoiding thermal runaway type failures. More specifically, SLID bonding requires melting of Sn during the bonding process and care should be taken to avoid Sn-squeeze out which would otherwise react with traces or RDLs to form

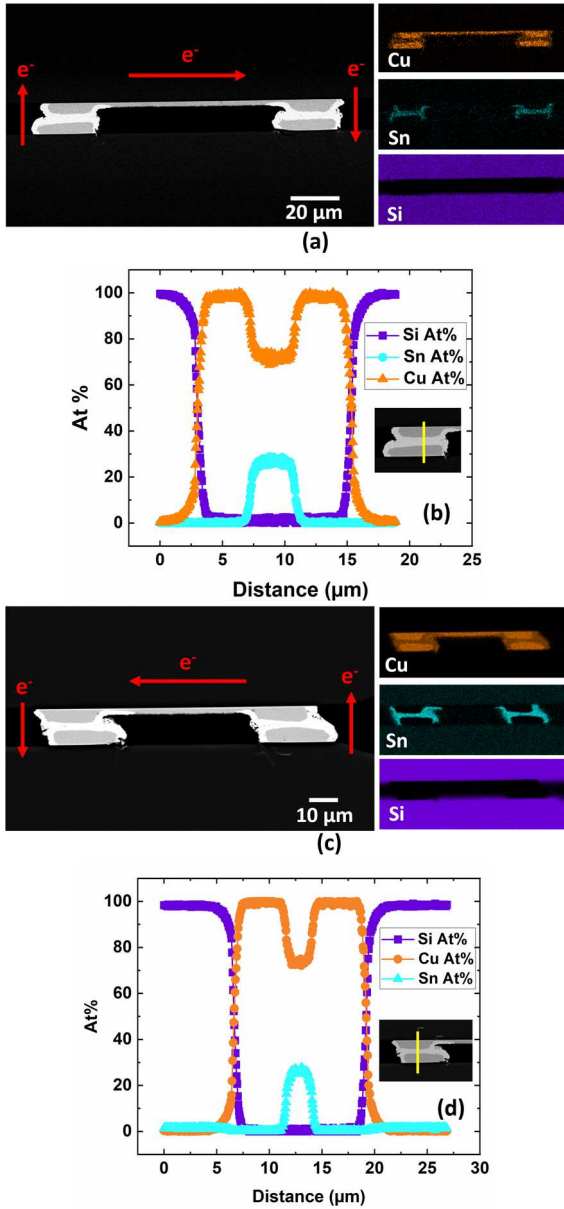


Fig. 8. (a) SEM image and elemental maps of 25 μ m TS1_B2 after the EM test. (b) atomic percentage versus distance plots of 25 μ m TS1_B2 after the EM test across cutline shown in the inset. (c) SEM image and elemental maps of 25 μ m TS2_B2 after the EM test, and (d) atomic percentage versus distance plots of 25 μ m TS2_B2 after the EM test across the cutline location shown in the inset.

IMCs at undesired locations. These locations in turn are the favorable sites for failures impacting the reliability and functionality of Cu₃Sn μ bumps. In batch-2, different sized μ bumps were incorporated in the mask to evaluate the Sn-squeeze out. Sn reacting with Cu tracks to form Cu₃Sn were still observed in the large sized μ bumps, Fig. 2(d), but not in the small sized μ bumps. Therefore, uniform dimensions of μ bumps across the wafer with optimized Cu, Sn electroplated stacks and bonding force could mitigate the negative effects of Sn squeeze-out in the stacked configuration.

E. Finite Element (FE) Simulations

FE simulations were carried out to understand the enhanced EM reliability of the 2-bump TS with Cu₃Sn μ bumps in batch-2 as compared to those from batch-1. Specifically, the current density distribution, temperature, and temperature gradient in the test structures from both the

batches were evaluated. The simulations were carried out in COMSOL Multiphysics utilizing joule heating physics in the heat transfer module. The 2-bump test structures were built mimicking the observations made from the cross-sections of samples from batch-1 and batch-2. Fig. 9(a-c) shows the image of the CAD model of the chip with a close-up view of TS without (ideal) and with Sn squeeze-out (non-ideal) effects. Fig. 9(d) shows the location of the cutline (1 μ m) below the surface of the Cu trace across which cutline graphs for different parameters are evaluated. The electrical conductivity of Cu and Cu₃Sn were assigned as 58.1×10^6 and 11.2×10^6 S/m respectively. The EM stress test conditions were incorporated for a current density of 1×10^5 A/cm² (current 2.5 A for 50 μ m TS) and at a temperature of 150 $^{\circ}$ C which was applied at the bottom surface of the bottom silicon chip to mimic the experimental conditions. Convective heat flux with a heat transfer coefficient of 5 W/(m².K) was assigned to the top surface of the top silicon chip. Fig. 9(e) shows the temperature evaluated across the cutline shown in Fig. 9(d). As can be seen, temperature of more than 160-180 $^{\circ}$ C (higher for non-ideal) could easily reach at the upper Cu trace because of joule heating even though the tests were done at 150 $^{\circ}$ C. Fig. 9(f) shows the temperature gradient profile and respective cutline graphs for non-ideal and ideal TS. Due to Sn squeeze out and formation of IMC at the Cu trace near the μ bumps, temperature gradient of an order of magnitude higher in the non-ideal case is observed as compared to the ideal case. These locations are also where current crowding and joule heating occurs. Thus, these factors combined have a detrimental effect on the EM reliability and functionality of IMC μ bumps at high current densities. Cu₃Sn with a high thermal stability (M.P. 676 $^{\circ}$ C) is observed to undergo thermal runaway and local melting at those locations, thereby highlighting the significance of proper design engineering of SLID μ bumps.

IV. CONCLUSION

In this work, EM reliability of Cu₃Sn IMC μ bumps manufactured *via* Cu-Sn SLID wafer-level bonding was investigated. The ill-effects of Sn squeeze-out and formation of Cu₃Sn at unwanted locations in the TS on its EM reliability and functionality were scrutinized. Non-destructive techniques such as TDR and LIT were employed to understand the location of probable failures after the EM tests. Subsequently, destructive analysis was carried out by preparing cross-sections of the TS for detailed investigation of the failures employing SEM and EDS. Significant improvement in the EM reliability of Cu₃Sn μ bumps manufactured without the adverse effect of Sn squeeze-out was observed. Finally, FE simulations were carried out for comprehensive understanding of the reasons for early failures in Cu₃Sn IMC μ bumps manufactured with adverse effects of Sn squeeze-out.

The Cu₃Sn IMC μ bumps are extremely resistant to electromigration provided the negative effects of Sn squeeze-out during the SLID bonding process is mitigated. The formation of Cu₃Sn IMCs at unwanted locations due to Sn-squeeze out could easily compromise the reliability of the Cu₃Sn μ bump resulting in failures, if the dimensions of the μ bump together with the electroplated Cu, Sn thickness and bonding parameters are not optimal. However, with proper design and optimized process parameters, SLID bonding technology offers several advantages in 3D HI and advanced packaging applications.

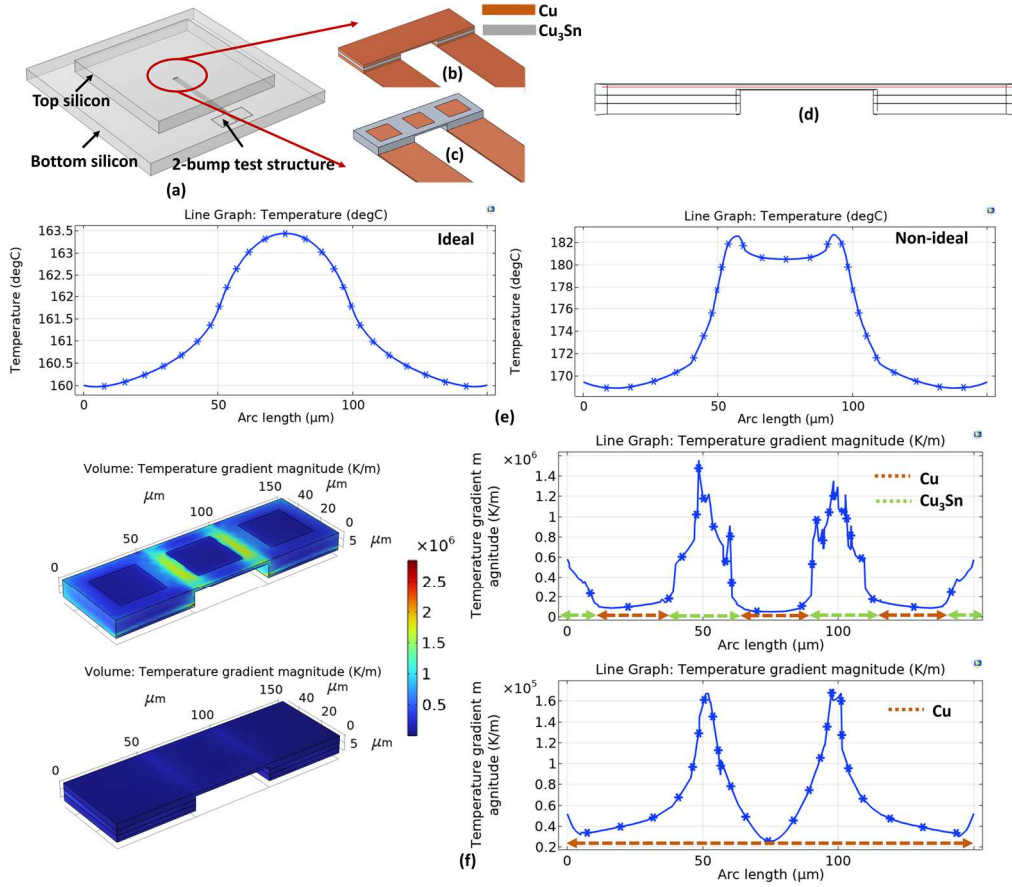


Fig. 9. (a) CAD image of the model. Close-up view of 2-bump TS in (b) ideal (w/o squeeze-out), and (c) non-ideal (with squeeze-out). (d) Cutline position in the model. (e) Temperature across the cutline shown in (d) in the ideal and non-ideal TS, and (f) temperature gradient plot and cutline graphs of non-ideal and ideal TS across the cutline shown in (d).

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REFERENCES

- [1] "IEEE international roadmap for devices and systems," 2020 ed., *Packag. Integr. White Paper*, 2017, pp. 1–22.
- [2] "IEEE heterogeneous integration roadmap," in *Chapter 1: HIR Overview and Executive Summary*, 2019 ed., 2019, pp. 1–14.
- [3] K. N. Tu, Y. Liu, and M. Li, "Effect of Joule heating and current crowding on electromigration in mobile technology," *Appl. Phys. Rev.*, vol. 4, 011101, 2017.
- [4] S. A. Chew, J. D. Vos, E. Beyne, "Wafer-to-wafer hybrid bonding at 400-nm interconnect pitch," *Nat. Rev. Electr. Engg.*, vol. 1, pp. 71–72, 2024.
- [5] J. H. Lau, "Recent advances and trends in Cu-Cu hybrid bonding," *IEEE Trans. Compon. Packag. Manuf. Technol.*, vol. 13, no. 3, pp. 399–425, 2023.
- [6] J. H. Lau, "Recent advances and new trends in flip chip technology," *J. Electron. Packag.*, vol. 138, no. 3, 030802 (23 pp), 2016.
- [7] J. H. Lau, "Recent advances and trends in advanced packaging," *IEEE Trans. Compon. Packag. Manuf. Technol.*, vol. 12, no. 2, pp. 228–252, 2022.
- [8] H.-Y. Chen *et al.*, "Electromigration immortality of purely intermetallic micro-bump for 3D integration," in *Proc. IEEE 65th Electron. Compon. Technol. Conf. (ECTC)*, San Diego, CA, USA, May 2015.
- [9] N. Tiwary, G. Ross, V. Vuorinen, M. Paulasto-Kröckel, "Impact of inherent design limitations for Cu-Sn SLID microbumps on its electromigration reliability for 3D ICs," *IEEE Trans. Electron. Dev.*, vol. 70, no. 1, pp. 222–229, 2023.
- [10] Y.-W. Chang *et al.*, "Electromigration mechanism of failure in flip-chip solder joints based on discrete void formation," *Sci. Rep.*, vol. 7, Art. No. 17950, 2017.
- [11] A. Munding, H. Hubner, A. Kaiser, S. Penka, P. Benkart, and E. Kohn, "Cu/Sn solid-liquid interdiffusion bonding," in *Wafer Level 3-D ICs Process Technology*, C. S. Tan, R. J. Gutmann, and L. R. Reif, Eds. New York, NY, USA: Springer-Verlag, 2008, pp. 131–169.
- [12] Y. Shi *et al.*, "Seed layer etching, thermal reflow and bonding of Cu-Sn micro bumps with 5 μm diameters," *Proc. IEEE 73rd Electron. Compon. Technol. Conf. (ECTC)*, Orlando, FL, USA, May 2023.